

Task 1 — Install Snort and verify the version

Step 1: Install Snort on Kali

Open the Kali terminal:

- `sudo apt update`
- `sudo apt install snort -y`

Step 2: Check the version

- `snort -V`

Step 3: Find your network interface and IP address

- `ip -br addr`

Task 2 — Monitor traffic and save alerts to snort.log

Step 1: Create a lab directory

- `mkdir -p ~/snort-lab`
- `cd ~/snort-lab`

Step 2: Create a basic scan-detection rule

Open a new rules file:

- `vim local.rules`

Paste this rule:

- `alert tcp any any -> any any (flags:S; detection_filter:track by_src, count 20, seconds 10; msg:"Possible TCP SYN scan"; sid:1000001; rev:1;)`

Step 3: Locate Snort's configuration file

Run:

- `ls -l /etc/snort/snort.lua`

Step 4: Start Snort

First, set the configuration path and interface. For example:

- `IFACE=eth0`
- `CONF=/etc/snort/snort.lua`

Then validate the configuration and rule:

- `sudo snort -c "$CONF" -R "$HOME/snort-lab/local.rules" -i "$IFACE" -T`

Task 3 — Detect an Nmap scan

Step 1: Find the Kali VM's IP

On Kali:

- `ip -4 addr show eth0`

Step 2: Run Nmap from the second VM

On your authorized scanning VM:

- `nmap -sS -p 1-1000 192.168.29.100`

Step 3: Check the Snort log

Back on Kali:

- `sudo grep -i "Possible TCP SYN scan" ~/snort-lab/snort.log`

If the rule triggers, you should see an alert containing:

- Possible TCP SYN scan

Task 4 — Alert only on HTTP traffic (port 80)

Step 1: Create the HTTP rule

On Kali:

- `cd ~/snort-lab`
- `vim http.rules`

Paste:

- `alert tcp any any -> any 80 (msg:"HTTP traffic detected";
sid:1000002; rev:1;)`

Save and exit.

Step 2: Validate the rule

- `sudo snort -c /etc/snort/snort.lua -R ~/snort-lab/http.rules -i eth0 -T`

Step 3: Start HTTP-only monitoring

- `cd ~/snort-lab`
- `sudo snort -q -c /etc/snort/snort.lua -R ~/snort-lab/http.rules -i eth0 -A
alert_fast > snort.log 2>&1`

Step 4: Generate HTTP traffic

Open a second Kali terminal and run:

- `curl -v --max-time 15 http://neverssl.com/`

Step 5: Verify the alerts

- `sudo grep "HTTP traffic detected" ~/snort-lab/snort.log`

To check for alerts mentioning other destination ports, inspect the complete log:

- `sudo cat ~/snort-lab/snort.log`